INCH-POUND
MIL-DTL-24643/59A
28 January 2010
SUPERSEDING
MIL-DTL-24643/59
9 September 2005

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, TYPE LSC5FS (INCLUDING VARIATIONS LSC5FSW, LSC5OS, AND LSC5OSW)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24643.

This cable can be manufactured in four variations: LSC5FS (Metallic Foil Shield, Non-Watertight), LSC5FSW (Metallic Foil Shield, Watertight), LSC5OS (Metallic Foil and Braid Shield, Non-Watertight), and LSC5OSW (Metallic Foil and Braid Shield, Watertight).

Construction (Metallic Foil Shield, Watertight and Non-Watertight)

First	-	Copper conductor, solid, 24 AWG nominal.
Second	-	Polyolefin insulation. Wall thickness as necessary, in order for the completed cable to comply with the specified electrical requirements. Maximum insulation diameter shall not exceed 0.046 inch.
Third	-	Two insulated conductors, twisted together such that they comply with the electrical requirements specified herein. Wire identification codes shall be in compliance with ANSI/TIA-568-C.2 as follows: Pair 1 – white/blue and blue Pair 2 – white/orange and orange Pair 3 – white/green and green Pair 4 – white/brown and brown
Fourth	-	Four pairs cabled together with a lay as required to comply with electrical requirements.
Fifth	-	For watertight variant, filler material shall comply with the requirements of 3.3.5 and 3.3.5.3 of MIL-DTL-24643.

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Construction Variant for Types LSC5FS and LSC5FSW

Sixth	-	An optional binder.
Seventh	-	24 AWG nominal TC drain wire.
Eighth	-	Metallic foil/polyester tape (metallic foil side facing in and in contact with the drain wire).
Ninth	-	Cross-linked polyolefin jacket. Minimum thickness of 0.040 inch.

Construction Variant for Types LSC5OS and LSC5OSW

Sixth	-	An optional binder.
Seventh	-	Metallic foil /polyester tape (metallic foil side facing out and in contact with the braid).
Eighth	-	Braided shield of 36 AWG or 34 AWG tin-coated copper. Coverage as required to comply with shield performance requirements.
Ninth	-	Optional binder.
Tenth	-	Cross-linked polyolefin jacket. Minimum wall thickness of 0.040 inch.

TABLE I. Details.

Military part no. M24643/59	**Type**	**Number of pairs**	**Overall diameter (in.)**	
			Min.	**Max.**
-01UO	LSC5FS-4	4	0.240	0.360
-02UO	LSC5FSW-4	4	0.240	0.380
-03UO	LSC5OS-4	4	0.290	0.400
-04UO	LSC5OSW-4	4	0.300	0.420

REQUIREMENTS:

Qualification Required.

INSPECTION:

Basic Electricals:

DC resistance (ohms/100 m, max.)	9.38
DC resistance unbalance (percent, max.)	5
Jacket flaws	No failure
Conductor and shield continuity	No failure
Voltage withstand	
Conductor to conductor	1.5kVAC or 2.5kVDC
Conductor to shield	1.5kVAC or 2.5kVDC

Group A:

Visual and dimensional	No failure

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Watertightness (Types LSC5FSW and LSC5OSW only)	No failure
Capacitance	
Mutual capacitance of a pair (nominal @1 kHz)	5.6 nf/100 m
Pair to ground capacitance unbalance (max. @1 kHz)	330 pf/100 m
Input impedance (1 MHz to 100 MHz)	100 ohms ±15%

TABLE II. Swept electrical parameters.

Frequency (MHz)	**1.0**	**4.0**	**10**	**16**	**31.25**	**62.5**	**100**
Return loss dB/100 m (min.)	20.0	23.0	25.0	25.0	23.6	21.5	20.1
Insertion loss dB/100 m (max.)	2.0	4.1	6.5	8.2	11.7	17.0	22.0
NEXT dB/100 m (min.)	65.3	56.3	50.3	47.2	42.9	38.4	35.3
ELFEXT dB/100 m (min.)	63.8	51.8	43.8	39.7	33.9	27.9	23.8
PS NEXT dB/100 m (min.)	62.3	53.3	47.3	44.2	39.9	35.4	32.3
PS ELFEXT dB/100 m (min.)	60.8	48.8	40.8	36.7	30.9	24.9	20.8
Propagation delay ns/100 m (max.)	570	552	545	543	540	539	538
Delay skew ns/100 m (max.)	45	45	45	45	45	45	45
NOTE: Values are for reference only. Actual values shall be determined utilizing the formulas in ANSI/TIA-568-C.2.							

Group B:

Drip – 95±1 °C (Watertight only)	Zero
Tear (lb/in thickness, min.) (ASTM D470)	35
Physicals (unaged)	
Insulation	
Tensile strength (lb/in^2, min.)	700
Elongation (percent, min.)	160
Jacket	
Tensile strength (lb/in^2, min.)	1300

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Elongation (percent, min.)	160
Thermoset proof test (percent, max.)	
Jacket	50
Group C:	
Physicals (aged air oven)	
Insulation (polyolefin)	
Tensile strength (percent of unaged, min.) (lb/in^2, min.)	75
Elongation (percent of unaged, min.)	75
Jacket (136 ºC for 168 hrs.)	
Tensile strength (percent retained, min.) (lb/in^2, min.)	60
Elongation (percent retained, min.)	60
Shrinkage	No failure
Cable filler removability (watertight only)	No failure
Heat distortion (percent of unaged, max.)	30
Conformance to material, construction and coverage	No failure
Surface transfer impedance (type LSC5OS and LSC5OSW only)	
Milliohms per meter, max.	700
EMP response, dB, min.	60
Permanence of printing (jacket) (cycles, min.)	125
Group D:	
Flame propagation (cable)	No failure
Qualification Inspection:	
Qualification inspection shall include basic electricals; groups A, B, C, and D; plus the following:	
Aging and compatibility (cable) 95±2 °C	No failure
Abrasion resistance (jacket)	75
Acid gas equivalent (percent, max.)	
Jacket	2
Fillers	8
Insulation	18
Halogen content (percent, max.)	
Jacket	0.2
Fillers	0.2
Insulation	0.2

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Immersion (jacket)	
Tensile strength (percent of unaged, min.)	50
Elongation (percent of unaged, min.)	50
Smoke index, max.	
Jacket	25
Fillers	55
Insulation	45
Toxicity index, max.	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) – Type A (hardness, min.)	80
Weathering (jacket)	No failure
Stress resistance	No failure
Temperature humidity (Types LSC5FSW & LSC5OSW only)	No failure

UNIT ORDERING LENGTHS:

Type and size	Feet (nominal)
LSC5FS, LSC5FSW, LSC5OS, LSC5OSW	500

CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army – MI
Navy – SH

Preparing Activity:
Navy – SH
(Project 6145-2009-102)

Review Activities:
Army – AR, AV, CR
Navy – CG, EC
DLA – CC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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